

CoRoT-1b

R-2345

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-1_221216 · created 2026-08-19T00:44:49+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459929.7455 +/- 0.0056 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.142 +/- 0.041
Transit depth (Rp/Rs)^2	2.02 +/- 1.16 [%]
Orbital Inclination (inc)	89.2 +/- 2.8
Airmass coefficient 1 (a1)	1.01 +/- 0.13
Airmass coefficient 2 (a2)	0.002 +/- 0.058
Scatter in the residuals of the lightcurve fit is	13.19 %
Best Comparison Star	#3 - [73, 136]
Optimal Aperture	2.77
Optimal Annulus	10.92
Transit Duration (day)	0.111 +/- 0.0057

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.142 ± 0.041 vs 0.1388 (deviation 0.07σ)
Duration fit vs published	0.111 d vs 0.1046 d (deviation 1.07σ)
Mid-time O–C	-4.9 min
Depth SNR	3.3
Residual scatter	13.19 %

Light curve

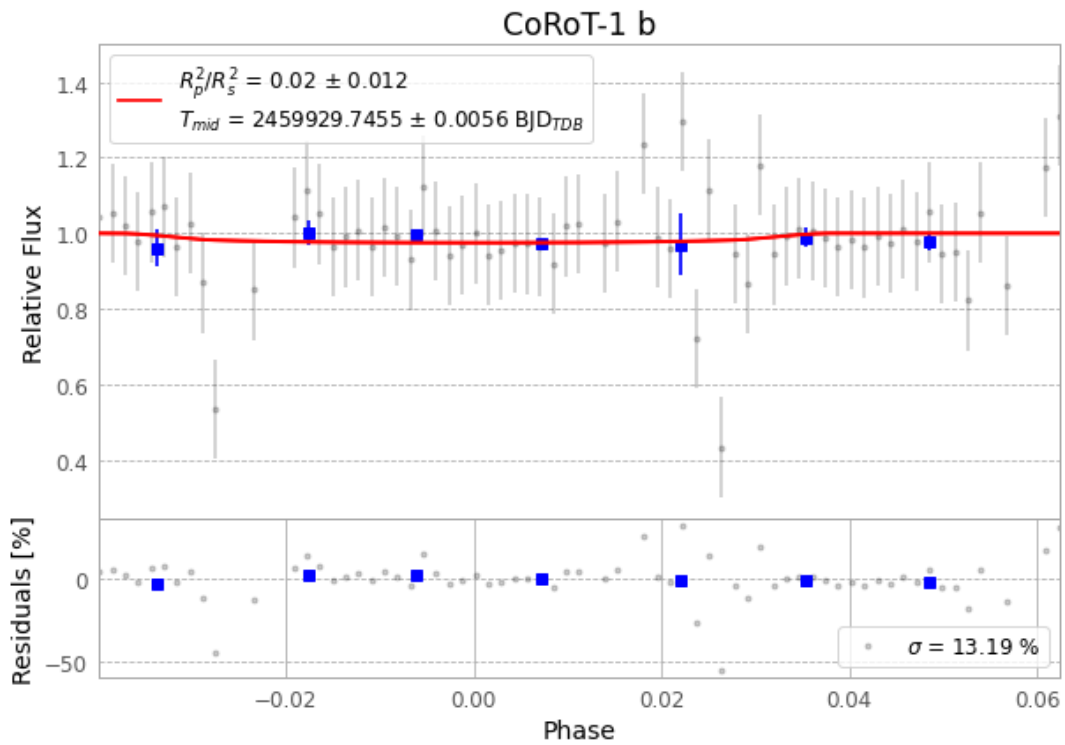


Figure 1 — FinalLightCurve_CoRoT-1 b_2022-12-15.png (SHA-256 62948bf47779db0b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [328, 35], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7d54aae168d88fb1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

76 FITS frames (50 MB), CoRoT-1221216041815.FITS → CoRoT-1221216080315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-1-221216.log (a734fa5d4e19...), FinalLightCurve_CoRoT-1 b_2022-12-15.png (62948bf47779...), FinalLightCurve_CoRoT-1 b_2022-12-15.csv (c3fc46292612...)

Compute resources

Wall time 179.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-19 00:44Z.